

David Dwyer

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EDUCATION

Texas A&M University

General Engineering – Bachelor of Science in Mechanical Engineering (Intended)

Aug 2026 – May 2030

College Station, TX

PROJECTS

Custom Go-Kart | *Frame Fabrication, Powertrain, Aluminum Fabrication*

Feb 2025

- Fabricated a go-kart frame from scratch around a 6.6 HP (224cc) Predator engine and custom-fitted CVT
- Machined engine mounting brackets from scrap aluminum sheet to work around tight chassis packaging
- Tuned powertrain performance by upgrading the exhaust and fitting a new carburetor for optimal air-fuel ratio
- Integrated chain drive, drive axles, mechanical brakes, throttle/brake cables, and a butterfly steering wheel
- Validated a 50 MPH top speed, then engineered a roll cage and scoped suspension for the next iteration

Scorpius – FTC Competition Robot | *Fusion 360, CNC Machining, TPU 3D Printing*

Aug 2025 – Jun 2026

- Modeled full robot subassemblies in Fusion 360, running clearance and stress checks before release to production
- Prototyped a fully modular robot architecture with interchangeable subsystems for rapid repair and iteration
- Specialized in compliant features 3D-printed from TPU, tuning part geometry for controlled flex and durability
- Machined structural components on a 3-axis CNC with hand and power tools to strict competition tolerances
- Integrated carbon fiber, CNC-machined aluminum, and 3D-printed parts into a lightweight competition chassis

HawkBot – FTC Competition Robot | *Fusion 360, CAD, Manufacturing*

Aug 2024 – May 2025

- Designed and iterated intake, lift, and drivetrain mechanisms in CAD to optimize weight distribution and efficiency
- Manufactured hybrid assemblies combining precision-machined custom parts with off-the-shelf hardware

TECHNICAL SKILLS

CAD / CAM: Onshape, Fusion 360, Computer-Aided Manufacturing (CAM), System Schematics

Fabrication: 3-Axis CNC Machining, 3D Printing, Carbon Fiber Layup, Aluminum Fabrication, Hand & Power Tools

Languages & Software: Java, Python, Microsoft Office, Google Suite

Certifications: American Lifeguard Association Lifeguard & AED/CPR Certified

LEADERSHIP / EXTRACURRICULAR

FIRST Tech Challenge Robotics – Teams 6832 & 3734

Sep 2022 – May 2026

Team Captain & Design Lead (2024–2026), Member (2022–2024)

Dallas, TX

- Led a 15-member team to Regional and State Championships twice across design, build, code, and outreach
- Directed design of 3 robots totaling 600+ custom 3D-printed, carbon fiber, aluminum, and CNC-machined parts
- Designed and released an open-source chassis architecture actively adopted by 12+ FTC robotics teams worldwide
- Earned **1st Place Design Award** (2025, 2026), **1st Place Sustain Award**, and **2nd Place Inspire Award**
- Secured ~\$13,000 in industry sponsorships by pitching corporate partners including Southwest Airlines

Boy Scouts of America – Troop 838

Apr 2025

Eagle Scout, Project Manager & Instructor

Dallas, TX

- Engineered and constructed rainwater collection platforms and navigation infrastructure for the community
- Managed a \$2,100+ budget and directed 25+ volunteers from planning through construction and install
- Trained younger scouts in field safety, technical skills, and leadership; project featured in Dallas news

WORK EXPERIENCE

Glen Cove Swim Club

Apr 2025 – Present

Pool Technician, Shift Leader, Lifeguard

Dallas, TX

- Diagnosed and repaired filtration, chemical balancing, and safety equipment to reduce facility downtime
- Supervised facility staff during daily operations, enforcing safety standards and resolving conflicts
- Earned **Rookie Lifeguard of the Year (2025)** for exceptional leadership and team coordination